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ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced)-2016

ANSWERS, HINTS & SOLUTIONS
HALF COURSE TEST–VII
(Paper - 2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**C**	**B**	**B**
2.	**B**	**A**	**A**
3.	**A, C**	**D**	**A**
4.	**A**	**C**	**B**
5.	**C**	**B**	**A**
6.	**A**	**B**	**C**
7.	**B**	**C**	**A**
8.	**A**	**B, C, D**	**A, B, C, D**
9.	**A, C, D**	**B, C, D**	**A, B**
10.	**A, C, D**	**A, D**	**A, B, C**
11.	**C**	**B, C**	**A, C**
12.	**C**	**D**	**C**
13.	**A**	**D**	**D**
14.	**B**	**A**	**A**
15.	**A**	**C**	**A**
16.	**C**	**D**	**B**
1.	**(A → s); (B → r); (C → q); (D → p)**	**(A → p, q, r, t); (B → p, q, t); (C → p, q, s, t); (D → p, q)**	**(A → s); (B → q); (C → r); (D → p)**
2.	**(A → q); (B → r); (C → t); (D → r)**	**(A → p); (B → t); (C → q, r, t); (D → p)**	**(A → s); (B → r); (C → r); (D → s)**
1.	**1**	**4**	**3**
2.	**2**	**2**	**4**
3.	**1**	**5**	**1**
4.	**4**	**3**	**7**
5.	**5**	**1**	**1**
6.	**3**	**1**	**1**

Physics

PART – I

Hints & Solution

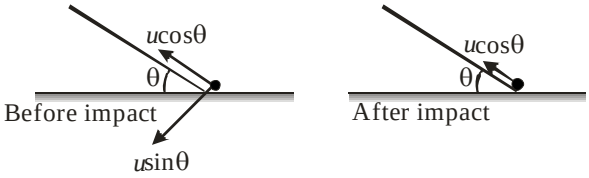
SECTION – A

1. $V_P \cos\alpha = u\cos\theta$

$$t = \frac{u\sin\theta - V_P\sin\alpha}{g} = u\sin\theta - \frac{u\cos\theta\sin\alpha}{\cos\alpha}$$

$$= \frac{u\sin(\theta-\alpha)}{g\cos\alpha}$$

2. Just before the particle moves to inclined surface, we resolve its velocity along and normal to the plane.



For the trajectory of the particle to sharply change from the horizontal line to the inclined line, the impact of the particle with inclined plane should reduce the u sin θ component of velocity to zero. Hence the particle moves up the incline with speed u cos θ.
Hence as θ increases, the height to which the particle rises shall decrease.

3. By balancing force along horizontal and vertical direction

$\mu N_1 \sin\theta + \mu N_2 \sin\theta = (N_1 - N_2)\cos\theta$ … (1)

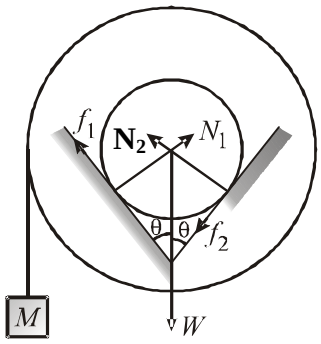
$(N_1 - N_2)\mu\cos\theta + (N_1 + N_2)\sin\theta = W + mg$ … (2)

Balancing torque about centre

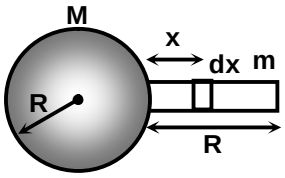
$(N_1 + N_2)\mu a = b\,mg$ … (3)

By solving (1), (2) and (3)

$$M = \frac{a\mu W}{(b(1+\mu^2)\sin\theta - \mu a)g}$$



4. $$F = \int_R^{2R} \frac{GM\left(\frac{m}{R}\right)dx}{x^2} = \frac{GMm}{2R^2}$$



5. $a_A = g\sin 37°$

Spring is at its natural length

Hence $a_B = g\sin 37° = 6\ m/s^2$

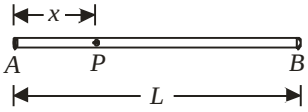
6. Total mass melted is 10 gm

Total heat in 100 sec 10 × 80

$$\text{Heat given to ice/sec} = \frac{800}{100} = 8\ \text{Cal/sec.}$$

$$\text{Rate of Rise of temp} = (10)(1)\ \frac{\Delta T}{\text{sec.}} = 8, \qquad \frac{\Delta T}{\text{sec.}} = .8°\text{C/sec.}$$

7. The KE of given section AP and PB will be equal if MI of each section AP and section PB about A is same.



$I_{AP} = (\lambda x)\frac{x^2}{3}$ $I_{PB} = I_{AB} - I_{AP} = (\lambda x)\frac{L^2}{3} - (\lambda x)\frac{x^2}{3}$

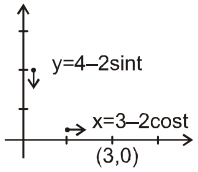
$\lambda I_{AP} = I_{PB}$ $\therefore \lambda\frac{x^3}{3} = \frac{\lambda L^3}{3} - \frac{\lambda x^3}{3}$

or $x^3 = \frac{L^3}{2}$ or $x = \frac{L}{2^{1/3}}$

8. If S is the distance between them at any time :

$\Rightarrow \quad S^2 = (4 - 2\sin t)^2 + (3 - 2\cos t)^2$

$= 29 - (16 \sin t + 12 \cos t)$ .... (1)



S will be 3 cm only when

$\Rightarrow \quad 16 \sin t + 12 \cos t = 20$

$\Rightarrow \quad 4 \sin t + 3 \cos t = 5$

$\Rightarrow \quad \frac{4}{5}\sin t + \frac{3}{5}\cos t = 1 \Rightarrow \sin(t + 37°) = 1$

Now the only solution is t=53° so only once in a time period, it will be 3 cm.
(B) By equation (1), S varies between 3 cm and 7 cm.
(C) Maximum magnitude of force is $\omega^2 A$.
(D) Kinetic energy is not conserved.

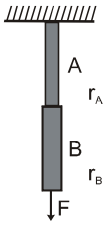
9. % Rise in area will be 0.4%
% Rise in vol will be 0.6%

$\left[1 + \frac{0.2}{100}\right] D = [1 + \alpha \times 80] D$

$\alpha = \frac{0.002}{80} = \frac{1}{400} = 0.25 \times 10^{-4} °C^{-1}$

10. Slope of displacement time graph is velocity and rate of change of velocity is acceleration.

11. That wire will break first whose stress reaches the breaking stress (for that material) first.



Stress = $\frac{F}{A} = \frac{F}{\pi r^2}$ where F is the tension in the wire. F is same in both the wires.

Thus stress $\propto \frac{1}{r^2}$ for the wire of smaller radius, the stress is more.

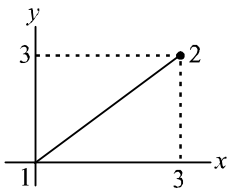
Thus the wire with smaller radius will break first.

If $r_A < r_B$, then wire A will break first.
If $r_A > r_B$, then wire B will break first.
If $r_A = r_B$, either of them could break.

12. $\vec{dl} = dx\,\hat{i} + dy\,\hat{j}$

$\vec{F} \cdot \vec{dl} = \alpha x y^2 dy$

on the path, x = y so $\vec{F} \cdot \vec{dl} = \alpha y^3 dy$



$W = \left(\frac{\alpha}{4}\right)\left|y^4\right|_{y_1}^{y_2} = \left(\frac{\alpha}{4}\right)\left(y_2^4 - y_1^4\right)$

$y_1 = 0$; $y_2 = 3.00$ m

So $W = \frac{1}{4}(2.50)(3)^4 = 50.6\,J$

(14-16). $Kx_0 + A\frac{L}{2}\rho g + Mg - 2Mg = 0$

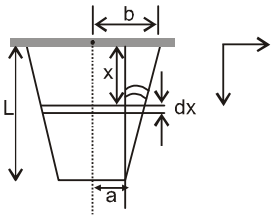
$K(x_0 + x) + A\left(\frac{L}{2} + x\right)\rho g + Mg - 2Mg = 3Ma$

$\omega^2 = \frac{g}{\text{amplitude}}$

SECTION – C

2. $Mg - T = Ma$
$T - mg = ma$
$T = \frac{2Mmg}{(M+m)}$
$\Rightarrow \frac{T}{A} = \frac{2Mmg}{A(M+m)} = \frac{2mg}{A\left(1+\frac{m}{M}\right)} = 2\times10^9$
$\Rightarrow$ M = 2

3. $Y = \frac{F/A}{dy/dx}, r = b - \left(\frac{b-a}{L}\right)x$
$r = b - \tan\theta x$, $A = \pi(b - x\tan\theta)^2$
$\int_0^{\Delta L} dy = \frac{F}{\pi Y}\int_0^L \frac{dx}{(b - x\tan\theta)^2}$
$\Delta L = 10^{-3}$m = 1mm

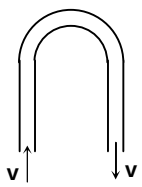


4. $|\varepsilon| = |\vec{v}\cdot(\vec{B}\times\vec{\ell})|$
$= (2\hat{i})\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 1 & -2\\ 0 & 3/5 & 4/5\end{vmatrix} = 4$

5. In fluid frame ball is moving with constant velocity (terminal) and resultant of all forces acting on it zero.
(A) Buoyant force
(B) viscous force
(C) mg

$v_t = \frac{2}{9}\frac{(\rho-\rho_0)gr^2}{\eta} = \frac{2}{9}\times\frac{900\times10}{0.1}\times25\times10^{-4}$ = 50 m/s

6. $2\rho Av^2 = 2F$
$\sigma = \frac{F}{3\pi r^2} = \frac{\rho A}{3\pi}$
$\sigma = \frac{\rho v^2}{3}$
v = 30 m/s



Chemistry

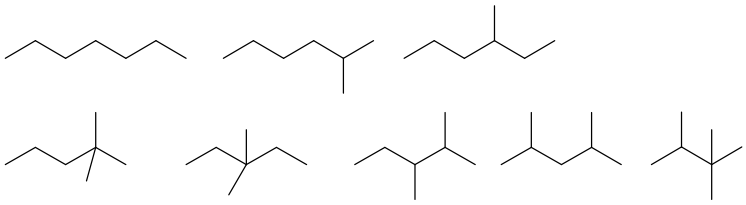
PART – II

SECTION – A

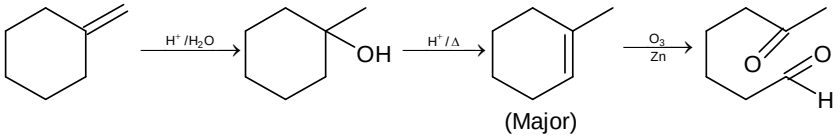
2. Strength of H – bond: 2 – 10 kcal or (8 – 42 kJ)

3. It may be possible that few particles of gas A have higher velocities than that of few particles of gas B.

4. 96 ppm of SO_4^{2-} = 100 ppm of $CaCO_3$ so 384 ppm of SO_4^{2-} = 400 ppm of $CaCO_3$

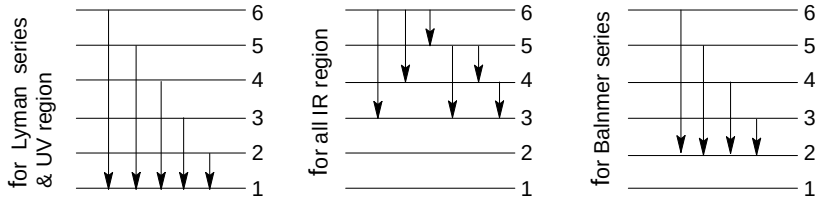
5. 

6. LiF is insoluble in water but not because of its covalent character rather its high lattice energy which is not easily compensated by its hydration energy.

7. 

(Major)

8. **Total lines**



9. All are extension of PV = nRT, $n = \frac{w}{M}$ & $d = \frac{w}{V}$

10. (B) and (C) are amphoteric.

12. The carbocation formed will be 3°, hence will be more stable.

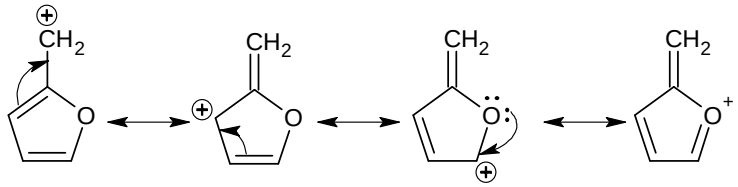
13. Compound (D) will form most stable carbocation on protonation.

14. All other molecules are symmetrical.

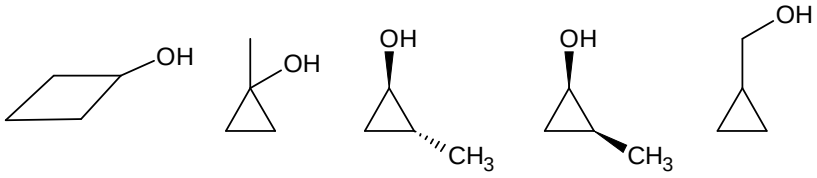
15. It contains only one lone pair so no chances of lp – lp repulsion.
In CO_2 & CH_4 there are no lone pairs and in XeF_2 two lone pair are there so lp – lp repulsion is also present.

16. Shape of I_3^- is linear despite of the presence of 3 lone pairs.

SECTION – C

1. 

2. In $Cu_2S(-2)$, In $FeS_2(-1)$, In $Na_2S_2O_3(+2)$
In $Na_2S_2O_4(+3)$

3. 

4. $A^{n+} \longrightarrow AO_3^-$
$x = +n \qquad x - 6 = -1 \quad \therefore x = 5$
$\therefore$ number of electron releases in oxidation = (5 – n)
$\therefore$ equivalent of $A^{n+} = 2 \times (5 - n)$
$\therefore$ equivalent of $M_nO_4^- = 0.8 \times 5$
$2 \times (5 - n) = 0.8 \times 5$
$n = 3$

5.

	A	+	2B	$\rightleftharpoons$	2C	+	D
	2		4		0		0
at eq.	2 – x		4 – 2x		2x		x

$2x = 2$ so $x = 1$

$$K_{eq} = \frac{(2x)^2 \times x}{(2-x)(4-2x)^2} = \frac{4}{4} = 1$$

6. Either $7 \to 1$ or $6 \to 1$ or $5 \to 1$ or $4 \to 1$ or $3 \to 1$ or $2 \to 1$
Only one transition can be obtained at a time as only single atom is given.

Mathematics

PART – III

SECTION – A

1. It represent 3 different non concurrent lines hence can represent a triangle

2. $\because$ $p < 0$ let $p = -q$, $q > 0$, so $p^{1/3} = -q^{1/3}(1)^{1/3}$
$\therefore$ $\alpha = -q^{1/3}$
$\beta = -q^{1/3}w$
$\gamma = -q^{1/3}w$
so given expression is $\dfrac{x + yw + zw^2}{xw + yw^2 + z} = \dfrac{1}{w} = w^2$

3. Let a, 3 be two number the
$A_1 + A_2 = a + b$
$G_1G_2 = \left(\sqrt{ab}\right)^2$
$\dfrac{1}{H_1} + \dfrac{1}{H_2} = \dfrac{1}{a} + \dfrac{1}{b} \Rightarrow \dfrac{H_1H_2}{H_1 + H_2} = \dfrac{G_1G_2}{A_1 + A_2}$

4. Mid point of the chord intercepted by hyperbola is same as mid point of the chord intercepted between asymptotes.
$\Rightarrow \dfrac{h+1}{2} = \dfrac{-2+4}{2}, \dfrac{k-2}{2} = \dfrac{-6+2}{2}$
$\Rightarrow (h, k) \equiv (1, -2)$.
$\Rightarrow$ line intersect asymptote at one point only hence it is the centre of hyperbola.

5. Let $y = \sin3A + \sin3B + \sin3C$ for sin3A to be non positive
we have $2\pi < 3A < 3\pi \Rightarrow \dfrac{2\pi}{3} < A < \pi$
since $A + B + C = \pi \Rightarrow$ all of sin3A, sin3B, sin3C can't be negative.
Let us take $\sin3A = -1 \Rightarrow A = \dfrac{\pi}{2} \Rightarrow \sin3A = -1$, sin3B = – 1 and sin3C = 0 is possible
$\Rightarrow$ the minimum value is – 2.

6. Sum of 7 digits = a multiple of 9
Sum of number 1, 2, 3, 4, 5, 6, 7, 8, 9 = 45 so two left should also have sum of 9.
The pair of left numbers is (1, 8), (2, 7), (3, 6), (4, 5).
with each pair left number of 7 digit number = 7!
So with all 4 pairs = $4 \times 7!$.

7. If p is a prime such that $100 > p > \dfrac{200}{3}$, then p and 2p but not 3p belong to [1, 2, …, 200].
While just p belongs to (1, 2, …, 100), so p does not divide $^{200}C_{100}$ but consider 61 the largest prime < $\dfrac{200}{3}$
61^3 divides 200! but only 61 divides 100!.
Hence 61 divides $^{200}C_{100}$.

8. In the expansion of $(a\alpha^2x^2 + 2b\alpha x + c)^n$
the sum of the coefficients = $(a\alpha^2 + 2b\alpha + c)^n$
Let $f(\alpha) = a\alpha^2 + 2b\alpha + c$
Its discriminant $= 4b^2 - 4ac = 4(b^2 - ac) < 0$
Hence, $f(\alpha) < 0$ or $f(\alpha) > 0$ for all $\alpha \in R$
If $a > 0$ then $f(\alpha) > 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)^n > 0$
If $c > 0$ i.e. $f(0) > 0 \Rightarrow f(\alpha) > 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)^n > 0$
If $a < 0$ then $f(\alpha) < 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)^n < 0$ if n is odd
If $c < 0$ i.e. $f(0) < 0 \Rightarrow f(\alpha) < 0 \Rightarrow (a\alpha^2 + 2b\alpha + c)^n > 0$ if n is even.

9. Roots of $x^2 + x + 1 = 0$ are complex cube roots of unity,
so $h(w) = h(w^2) = 0$
$\Rightarrow w f(1) + w^2 g(1) = 0$ and $w^2 f(1) + w g(1) = 0$
$\Rightarrow f(1) = g(1) = 0$.

10. $\dfrac{a+b+c}{3} > b\ (\text{H.M})$
$\Rightarrow a + c > 2b$
Also $\dfrac{a^2+c^2}{2} > \sqrt{a^2c^2} \Rightarrow a^2 + c^2 > 2ac > 2b^2$ (G.M. > H.M)
$\Rightarrow a^2 + c^2 > 2b^2$.

11. $\sqrt{y + \dfrac{1}{y}} \geq \sqrt{2}$ and $|\sin x + \cos x| \leq \sqrt{2}$.
Hence $y + \dfrac{1}{y} = 2$ and $\sin x + \cos x = \sqrt{2}$, which is possible for $y = 1$, $x = \pi/4$.

12. Solving it, we get $\sin(\alpha + \beta) = \sin\alpha \pm \sqrt{-\cos^2\beta} \Rightarrow \cos\beta = 0 \Rightarrow \beta = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$
(i) $\beta = \dfrac{\pi}{2} \Rightarrow \tan\alpha = 1, \alpha \in \left\{\dfrac{\pi}{4}, \dfrac{5\pi}{4}\right\}$.
(ii) $\beta = \dfrac{3\pi}{2} \Rightarrow \tan\alpha = -1, \alpha \in \left\{\dfrac{3\pi}{4}, \dfrac{7\pi}{4}\right\}$.

13. Let $\cos A + \cos B + k\cos C = y$
$\Rightarrow 2k\sin^2\dfrac{C}{2} - 2\cos\left(\dfrac{A-B}{2}\right)\sin\dfrac{C}{2} + y - k = 0$. As $D \geq 0$
$\Rightarrow k(y - k) \leq \dfrac{1}{2}\cos^2\left(\dfrac{A-B}{2}\right) \leq \dfrac{1}{2} \Rightarrow y \leq \dfrac{1}{2k} + k$.

14–16. Let the roots of $z^3 + az^2 + bz + c = 0$ be α, β, γ, then
$|c| = |\alpha\beta\gamma| = 1$ and $|\alpha\beta + \beta\gamma + \gamma\alpha| = \left|\dfrac{1}{\alpha} + \dfrac{1}{\beta} + \dfrac{1}{\gamma}\right|$
$= |\alpha + \beta + \gamma| = |a|$
$\Rightarrow |a| = |b|$ and $|c| = 1$
also, $|a| = |\alpha + \beta + \gamma| \leq 3$
put $|a| = 2k + 1$, where $-\dfrac{1}{2} \leq k \leq 1$
thus $z^3 + |a|z^2 + |b|z + |c| = z^3 + (2k + 1)z^2 + (2k + 1)z + 1$

$= (z + 1)(z^2 + 2kz + 1)$.

It has roots -1 and $-k \pm \sqrt{k^2 - 1}$ since $k^2 \leq 1$, we have $\sqrt{k^2 - 1} = i\sqrt{1 - k^2}$ so $\left|-k \pm \sqrt{k^2 - 1}\right|^2 = k^2 + 1 - k^2 = 1$.

so, all the roots have modulus 1.

SECTION – B

1 (A). $y^2 = 4x$

Clearly, if P $(at^2, 2at)$ then by symmetry,

Q $(at^2, -2at)$

Equation of tangent is

$ty = x + at^2$

for t, $y = 0$, $x_1 = -at^2$

and equation of normal is

$y = -tx + 2at + at^3$

for R, $y = 0$, $x_2 = (2a + at^2)$

Here, $a = 1 \Rightarrow x_1 = -t^2$ and $x_2 (2 + t^2)$

$x_2 = 3 = 2 + t^2 \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$

Take $t = 1$, then $x_1 = -1$, $\therefore$ PM $= 2at = 2 \times 1 \times 1 = 2$

$RT = x_1 + x_2 = (1 + 3) = 4$

$\therefore$ Area of quadrilateral PTQR $= 2 \times \left(\frac{1}{2} \times 4 \times 2\right) = 8$ sq. units.

(B) Length of latus rectum + tangent PT $= 4a + \sqrt{S_1}$

$= 4 \times 1 + \sqrt{0^2 - 4(-1)} = 4 + 2 = 6$ units.

(C) Clearly, RT will be the diameter of circle

$\therefore$ Circumference $= (\pi \times$ diameter$)$

$= \pi \times RT = \frac{\pi \times 4}{4\pi} = 1$.

(D) Since in the part (1), we have found $x_2 (2a + at^2) > 2a$, (if $t \neq 0$)

$\therefore$ For three real normals, $x_2 > 2a = 2 \times 1 = 2$

i.e. $x_2 > 2$.

2. (A) $s_p = \frac{p}{2}[2A + (p-1)d] = a$

$$\frac{2a}{p} = 2A + (p-1)d \quad \ldots (1)$$

$$\frac{2b}{q} = 2A + (q-1)d \quad \ldots (2)$$

$$\frac{2c}{r} = 2A + (r-1)d \quad \ldots (3)$$

Multiply (1), (2) and (3) by $(q - r)$, $(r - p)$ and $(p - q)$ and add

$\therefore \sum \frac{a}{p}(q-r) = 0$.

(B) Let common ratio be taken as k and a be the first term
$R = T_r = ak^{r-1}$
$R^{s-t} = a^{s-t}\, k^{(r-1)(s-t)}$
Similarly, $s^{t-r} = a^{t-r}\, k^{(s-1)(t-r)}$
$T^{r-x} = a^{r-s}\, k^{(t-1)(r-s)}$.
Multiplying the above three and knowing that
$A^m A^n A^p = A^{m+n+p}$
$R^{s-t}\, S^{t-r}\, T^{r-s} = a^0 k^0 = 1$.

(C) $x = a + (m-1)d = AR^{m-1}$
$y = a + (n-1)d = AR^{n-1}$
$z = a + (p+1)d = AR^{p-1}$
$y - z = (n-p)d,\ z - x = (p-m)d,\ x - y = (m-n)d$
$x^{y-z} \cdot y^{z-x} \cdot z^{x-y} = \left(AR^{m-1}\right)^{(n-p)d} \cdot \left(AR^{n-1}\right)^{(p-m)d} \cdot \left(AR^{p-1}\right)^{(m-n)d}$
$= A^0 R^0 = 1$.

(D) $a = AR^{p-1},\ b = AR^{q-1},\ c = AR^{r-1}$ for G.P.
$a = \dfrac{1}{A_1 + (p-1)D},\ b = \dfrac{1}{A_1 + (q-1)D},\ c = \dfrac{1}{A_1 + (r-1)D}$ for H.P.
$\dfrac{1}{b} - \dfrac{1}{c} = (q-r)D$ … (1)
$\log a = \log A + (p-1)\log R$
$= (\log A - \log R) + (\log R)$ … (2)
$\sum\left(\dfrac{1}{b} - \dfrac{1}{c}\right)\log a = D\left(\log A - \log R\right)\sum(q-r) + D\log R\sum p(q-r) = 0$
$\therefore\ \sum\dfrac{c-b}{bc}\log a = 0$
multiplying by – abc
$\sum a(b-c)\log a = 0$.

SECTION – C

1. Let a – d, d, a + d be roots then
$\left.\begin{matrix} a - d + d + a + d = 12 \\ (a-b)d(a+d) = 28 \end{matrix}\right\}$ solve $\begin{matrix} a = 4 \\ d = \pm 3 \end{matrix}$
$\therefore\quad |d| = 3$

2. Let root of $x^2 - 3x + c = 0$ be α, β and that of $x^2 - cx + b = 0$ is α', β'
$\alpha - \beta = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \sqrt{b^2 - 4c}$ r
$\alpha' - \beta' = \sqrt{c^2 - 4b}$ so $\alpha - \beta = \alpha' - \beta'$ gives (b + c)(b – c) = 4(c – b)
$\therefore$ – (b + c) = 4.

3. We know that $\dfrac{c_1}{c_0} + 2\dfrac{c_2}{c_1} + \ldots + n\dfrac{c_n}{c_m} = \dfrac{n(n+1)}{2}$.

6. For sum of coefficients put x = 1 and so we get $(\alpha - 1)^{35} = (1 - \alpha)^{35} \Rightarrow \alpha = 1$.